

Australian Dwelling Approvals Forecasting Report

Attribution

The data used in this report was sourced from:

Australian Bureau of Statistics. (2025). Building Approvals, Australia, June 2025. Available at: <https://www.abs.gov.au/statistics/industry/building-and-construction/building-approvals-australia/jun-2025>

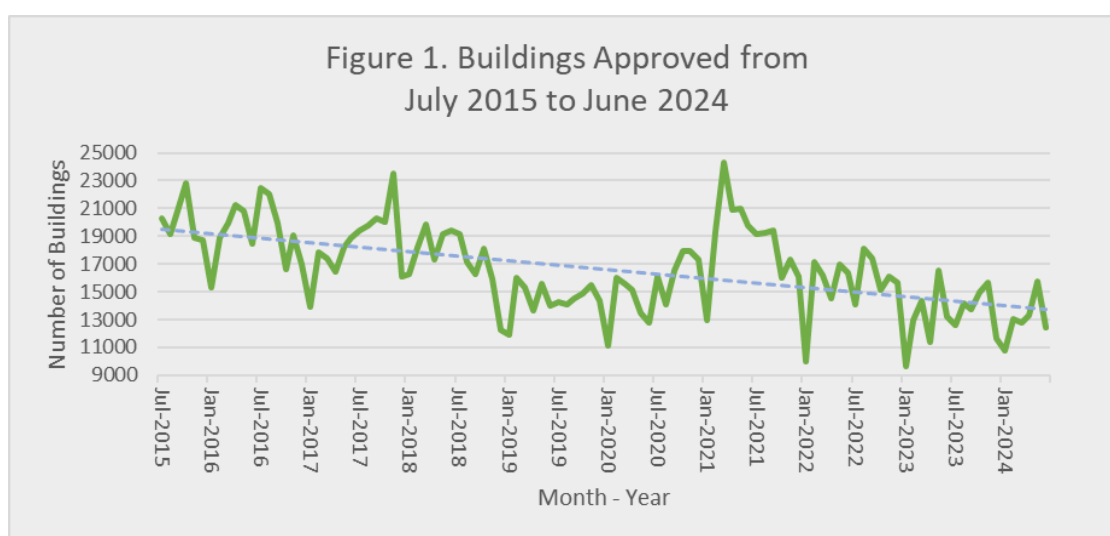
Scope

In this analysis, forecasts were made using relevant data from July 2015 to June 2024. This data contains a sequence of values on a variable collected at successive points in time, so it can be treated as a time series.

The aim is to forecast the number of buildings approved from July 2024 to June 2025 using the Winters' Exponential Smoothing model. This model is particularly useful for time series with level, trend and seasonal components. In a time series, level represents the underlying value of the series for a given time period; trend represents increase or decrease in the level and seasonality represents a repeating pattern occurring at fixed intervals. If the seasonal component is absolute, an additive model is appropriate. A multiplicative model is preferred otherwise. There are 3 parameters controlling the model's performance. Accordingly, their optimal values can be identified using the tool SOLVER in Excel.

Application

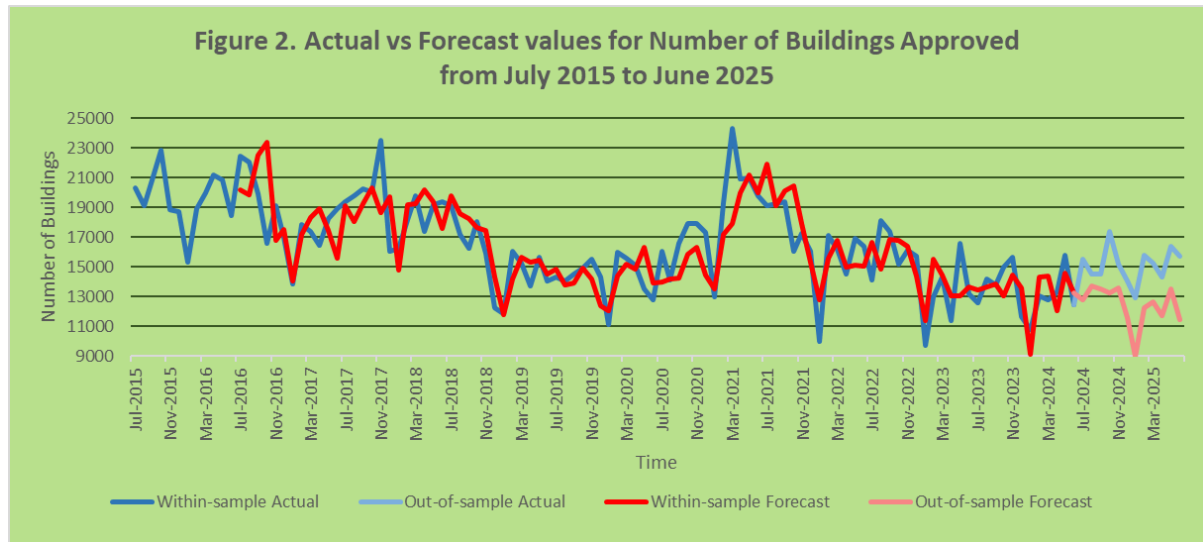
The graph below shows the number of buildings approved from July 2015 to June 2024, helping examine the pattern of the time series.



The line graph suggests the presence of level, trend, seasonal and random components. As the seasonal component does not look absolute over the time, a multiplicative model is more appropriate in this case. The seasonal period is 12 months long.

The initial value for L_t was estimated by averaging the first 12 values of the time series, T_t was found by averaging the monthly differences over the first two years. Finally, the first twelve S_t were obtained by dividing each of the first twelve actual values by the June 2016 value. Both within-sample and out-of-sample forecasts were then generated by applying relevant WES equations. The mean squared error (MSE) could be calculated after that. Using SOLVER in Excel, the optimal values for α , β and γ were found via the MSE. From here, forecast values also changed in response to the new parameter values.

Analysis



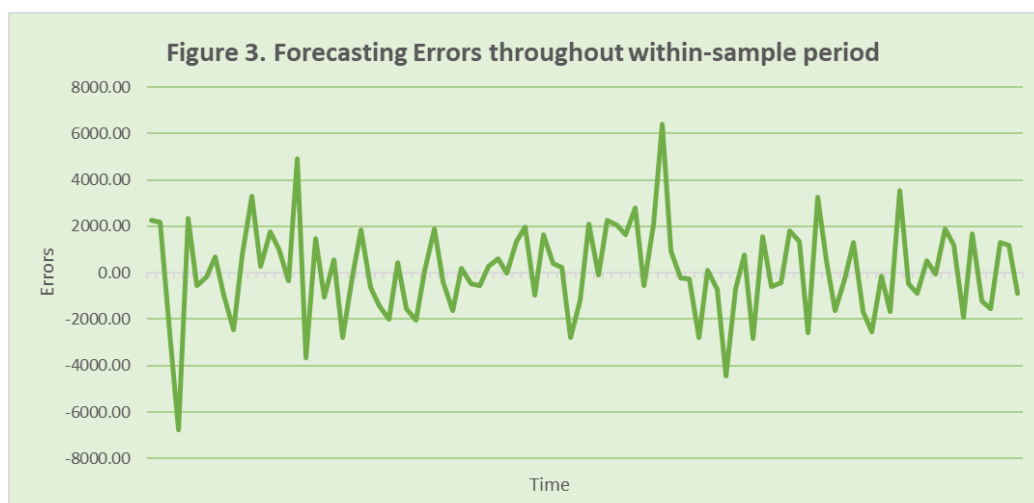
The line graph above was created using actual and forecast values from both within-sample and out-of-sample periods. The blue and red lines represent the actual and forecast values respectively. The lighter-coloured section of the lines indicates the out-of-sample values. During the within-sample period, the two lines move closely together, suggesting that the model fits the in-sample data well. During the out-of-sample period, although the two lines tend to move in different directions, they still follow the same pattern, suggesting that the out-of-sample forecasts are reasonably reliable and still explain the variation in unseen data to some extent.

From what Figure 2 suggests, this multiplicative WES model works quite well. During the within-sample period, the number of buildings approved experienced a downward trend, making the model assume this trend when making out-of-sample forecasts. The out-of-sample data, unfortunately, did not closely follow this trend, resulting in larger differences between actual and forecast values. This may suggest the presence of cyclical components that the model does not fully capture.

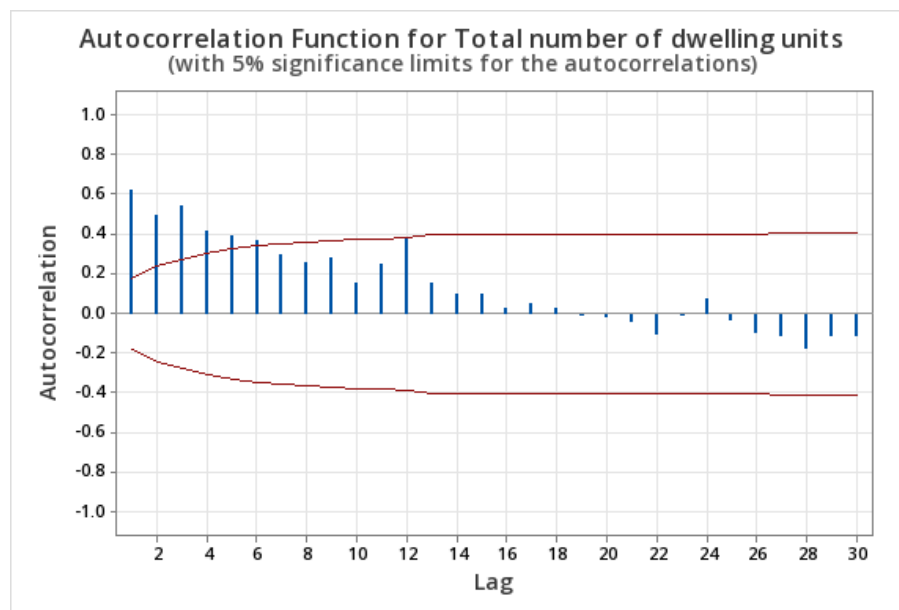
Articulation of Issues

Firstly, a correctly specified model should have smaller errors than other alternative models. Because the optimal parameter values for this model were obtained by minimizing the MSE, this specification is satisfied.

Secondly, the model should produce distributed errors with a zero mean. This graph supports this specification.



Finally, a correctly specified model should explain the systematic components of the generating process. However, the correlogram below shows moderate correlations at irregular lags, which may indicate the presence of components that the model does not adequately capture.



Critique

This post-optimised model is considered the best among all the WES models. As noted, the seasonal component is not constant, making the multiplicative model more appropriate than any additive models. Besides, this model produces more accurate forecasts than the pre-optimised model. The MSE of this model (3,818,644.27) is much lower than that of the pre-optimised one (5,156,169.93). However, the model is still not fully capable of capturing all components in the data, as certain factors remain unexplained.

These factors should be considered to help the company set reasonable expectations for forecasting performance. The most relevant factors influencing the number of buildings approved are economic ones, including interest rate, property prices and employment. Strong employment, low interest rate and rapidly growing housing prices tend to increase demand for housing and construction, thereby affecting the number of buildings approved. Other factors can also affect approvals. Some of the factors are predictable while the others are not. Therefore, the company must consider any potential factors to expect changes and adjust business plans in response to the unexpected factors.

Position

In conclusion, assuming future conditions remain similar to those in the past, the constructed model is reasonably reliable and useful for forecasting, offering higher accuracy than many other alternatives. The main limitation is that it fails to capture some of the more complex components in the data. The company should consider potential factors that may alter the overall pattern, allowing better anticipation of expected approval numbers.